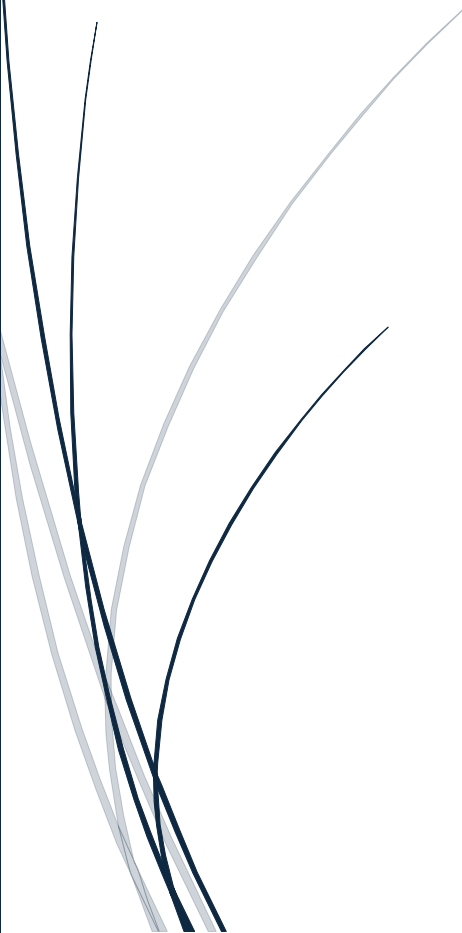




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Can Data analysis be used to predict crop yields?

MSc Project Proposal



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Introduction

Automation is increasing in all areas of industry and agriculture is no exception. With automated pickers increasing productivity and sensors giving us more information, data is helping drive key decisions. This growing reliance in these kinds of systems shows a new transformation in modern farming, where digital tools are being used to monitor crops and optimise resources. Advances in sensors now allow farmers to collect information about different factors, like soil health and local climate, continuously which allows for more precise management.

Aims and objectives

Aim

In this project, the aim is to investigate how data analysis can be used to predict crop yields, while ensuring data quality, selecting appropriate machine learning modelling methods and identifying the most important variables that contribute to accurate and reliable yield predictions.

Objectives

The objectives for the project are:

1. Assess the quality of free datasets to produce models
 - Examining freely available datasets that include information on factors, like weather and soil types, to see if they are complete, consistent and accurate
2. Compare and contrast different Machine Learning models
 - Select relevant machine learning models based on the literature
 - Test multiple machine learning models using standard metrics, for example mean square error, recall and accuracy
3. Identify Key variables for the construction of these models
 - Using standardised statistical measures, identify the main factors that affect crop yields, for example, backward elimination

Research Context

Recent literature shows there is more use of AI and remote sensors being used to address issues in productivity, food security and climate change across the globe. Across these three initial papers reviewed, three themes appeared:

- AI being used to improve accuracy in yield predictions
- Remote sensors being used for monitoring crop growth
- The need for deployable systems across fields to get interpretable data

The work of (Wakchaure, Patle and Mahindrakar, 2023) investigates how AI and robots can help improve different phases of agriculture. The authors investigate 12 different AI

techniques and see how each one is used across different simulations and experimentation. They also separate the farming process into the three phases of Cultivation, Monitoring and Harvesting to show where automation is used most and where it is still needed. Although the dissertation will be mainly focusing on monitoring, this article gives some insight into which AI methods are useful for tasks like crop health and yield predictions. It will also give some ideas into what techniques can be used to help predict crop yields from the different factors.

In the article by (Al-Adhaileh and Aldhyani, 2022), the authors look into how AI can be used to predict crop yields and why accurate forecasting is needed. This article gives some ideas about how machine learning (like SVM and Random Forest), deep learning (like CNN and LSTM) and neural network models can help predict the crop yields. The authors also give the benefits and limitations of AI yield predictions as well as some ideas of how to reduce the limitations.

In the article by Ilyas, Ahmad and Mehmood (2023), they show an AI system for crop type classification and crop yield estimation using remote sensors. This system uses inputs from satellite images and environmental sensors to collect real-time data. It applies fuzzy image to clean and improve the satellite images and then uses a CNN to differentiate between crop types. For yield prediction, the system trains machine learning models on historical FAO and World Bank datasets. The authors stated that by using this system they were able to increase crop classification accuracy and gradient boosting was the best at yield prediction.

The proposed research design

To achieve the aims and objectives, data will be required which focuses on crop yields and contains relevant data pertaining to the factors that affect the growth of the crop. These datasets will be sourced from open sources such as Kaggle, EcoCrop and Government websites. It will be verified that the data contained in the datasets are consistent to allow for fair comparison.

R and RStudio will then be used to perform analyses and data manipulations. I will also implement relevant machine learning techniques to find key factors and produce relevant graphical outputs.

Project Risk Assessment

Risk	Mitigation	Likelihood	Impact	Total Score (Likelihood x Impact)	Likelihood after Mitigation	Impact after Mitigation	Total Score After Mitigation
No Relevant Information	- Check if information is available before starting - Find multiple Sources for data - Adjust research question if needed	2	4	8	1	3	3
Model not working	- Start with a simple model before trying others - Test to try and find issues - Use version control and save results - Ask peers for advice	3	4	12	2	4	8
Illness	- Add buffer time to my timeline - Prioritise the most important tasks - Keep documentation to continue sooner	2	3	6	2	2	4
Power Outage	- Save Work frequently/autosaves - Keep a laptop charged - Work in a location that has	1	3	3	1	2	2

	power • Store files in the cloud						
Computer Stops working	• Back up files to the cloud • Have software on another device • Use university computers	2	5	10	1	3	3
Losing backups	• Have project stored in multiple locations • Have version control	2	5	10	1	4	4
Software Issues	• Install updates early • Have similar software available	3	3	9	2	3	6
Internet Issues	• Download datasets • Work offline as much as possible • Use university facilities	2	3	6	2	2	4
Scope Creep	• Have clear objectives • Identify the main parts • Get advice before adding new tasks	2	4	8	1	3	3

Ethics

This project does not have any ethical concerns because it relies on publicly available datasets rather than collecting information from participants or using any sensitive information like from Kaggle or Government websites. All the datasets that will be used are openly accessible and can be used for research. The data used is non identifiable and does not contain personal or confidential information.

Impact

By improving how we model crop suitability and yields are affected by environmental conditions, this research has the potential to have impact which helps to contribute to global food security, supports planting crops during a changing climate and contributes to growth in the economy.

The United Nations have 17 sustainable development goals, with one being Zero Hunger (SDG 2). This involves achieving food security and promote sustainable agriculture. By looking at what environmental factors and how they affect the yields, we can help improve those factors, (like if a farm is highly prone to flooding, better drainage can be implemented) and therefore increase the yields which means more food can be available globally.

It can also help with environmental issues such as climate change and air quality (UNSG 13). Since plants absorb pollutants like carbon dioxide and release oxygen, more crops means that there is less pollutants in the atmosphere and impact on climate change would be reduced. If crops are planted efficiently, crops can also help the soil they grow on to improve without needing to rely on artificial means.

Also, by increasing crop yields, it can also promote economic growth. With more crops available, it would ensure that there is a stable food supply and food prices would be reduced. More crops could also be exported to other countries increasing the economy of the country.

References

Al-Adhaileh, M. H. and Aldhyani, T. H. H. (2022) 'Artificial intelligence framework for modeling and predicting crop yield to enhance food security in Saudi Arabia'. *PeerJ Computer Science*, 8, p. e1104. Available at: 10.7717/peerj-cs.1104.

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